Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/19/2017
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Evaluation of Brain White Matter Integrity in NASA Crewmembers: Comparisons with Controls, Changes with Flight, and Behavioral Consequences		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/01/2016	**End Date:**	03/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SA00003		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to address this important issue through the following aims: 1) We will access diffusion weighted images from the Lifetime Surveillance of Astronaut Health (LSAH) that have been collected on NASA astronauts pre and post flight over the past eight years (permission has been granted from Dr. Riascos and LSAH staff, potential n = 21). We will compare pre flight measures of white matter integrity to data that we have already collected in 16 controls, and we will quantify and localize any changes in white matter integrity from pre to post flight. Some of these crewmembers also have pre and post flight FLAIR MRI scans so we could quantify WMHs in these individuals and examine how they relate to diffusion weighted metrics. 2) We will test for correlations between white matter integrity changes in these crewmembers and behavioral changes in posture control, locomotion, and cognition. We have confirmed that several of these previous crewmembers have participated in Dr. Bloomberg's Functional Task Test (FTT) study, and pre and post flight balance measures have been collected as part of the medical monitoring process. We will also reach out to other		

	investigators; for example, Dr. Dinges, who may also have data quantifying cognitive and other physiological changes from pre to post flight on the same crewmembers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work is critical for understanding the effects of spaceflight on the human brain and as such serves as a risk assessment analysis. The results will help us to further understand multisensory and sensorimotor brain neuroplastic changes occurring on Earth as individuals adapt to unloading and altered sensory inputs that can occur with age, immobility, vestibular disorders, and other conditions.
Task Progress:	Recent reports of elevated brain white matter hyperintensity (WMH) counts and volume in post flight astronaut MRIs warrant further examination of the impact of spaceflight on brain white matter microstructure. Moreover, white matter plasticity may occur with the extensive sensorimotor adaptation that occurs with microgravity exposure. Thus, for this study we evaluated retrospective longitudinal diffusion weighted MRI scans obtained from 15 astronauts pre- and post-flight. In light of the recent reports of microgravity-induced cephalad fluid shift and gray matter atrophy seen in astronauts, we applied a technique to estimate white matter metrics after correcting for free water contamination. This approach enabled the analysis of white matter tissue-specific alterations that are unrelated to fluid shift, occurring from pre- to post- spaceflight as well as regional shifts in free water (FW). White matter fractional anisotropy (FA) values decreased in crewmembers from pre- to post-flight relative to ground based controls in an area encompassing the superior and inferior longitudinal fasciculi and the inferior fronto-occipital fasciculus. Increased radial diffusivity (RD) and decreased axial diffusivity (AD) were also detected within overlapping regions. In addition, FA values in the corticospinal tract decreased and RD measures in the precentral gyrus white matter increased from pre- to post-flight. The degree of change in DTI metrics and free water volume of the astronauts were greater than that observed in a normative control group who were also evaluated over a prolonged time interval. The results show disrupted white matter structural connectivity in tracts involved in visuospatial processing, vestibular function and movement control as a result of spaceflight. RD increases were associated with the number of cumulative days spent in microgravity and with greater balance declines from pre- to post-flight. These sensory and motor white matter tract declines may arise due to body unloading and altered vestibular and tactile inputs that occur in microgravity and likely contribute to the constellation of sensorimotor declines that are observed post-flight. Prospective longitudinal studies are required to evaluate the time course of recovery from these changes. We also conducted a pilot sample analysis in an effort to determine whether white matter changes were associated with pre-flight training or with microgravity exposure. Combining the retrospective diffusion weighted MRI (DWI) scans with our ongoing prospective flight study yielded three naïve flyers. We compared their DWI pre-flight scans to those of repeat flyers and to normative control subjects. There was no difference in white matter metrics between the naïve flyers and ground based control subjects. Repeat flyers had reduced FA and greater free water in several brain regions in comparison to the naïve flyers. Our cautious interpretation (due to the very small sample size) is that pre-flight training does not substantially affect white matter, whereas spaceflight does. Data analyses are nearly complete and we are drafting manuscripts. [Ed. Note: see Bibliography for published manuscript. (April 2020)] [Ed. Note: See Bibliography for published manuscript. (epub noted in Aug. 2021; updated to include pub info in Jan. 2024)]
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Articles in Peer-reviewed Journals	Lee JK, Koppelmans V, Riascos RF, Hasan KM, Pasternak O, Mulavara AP, Bloomberg JJ, Seidler RD. "Spaceflight-associated brain white matter microstructural changes and intracranial fluid redistribution." JAMA Neurol. 2019 Apr 1;76(4):412-9. https://doi.org/10.1001/jamaneurol.2018.4882 ; PubMed PMID: 30673793; PubMed Central PMCID: PMC6459132 , Apr-2019
Articles in Peer-reviewed Journals	Hupfeld KE, McGregor HR, Koppelmans V, Beltran NE, Kofman IS, De Dios YE, Riascos RF, Reuter-Lorenz PA, Wood SJ, Bloomberg JJ, Mulavara AP, Seidler RD. "Brain and behavioral evidence for reweighting of vestibular inputs with long-duration spaceflight." Cerebral Cortex. 2022 Feb 15;32(4):755-69. https://doi.org/10.1093/cercor/bhab239 ; PMID: 34416764 ; PMCID: PMC8841601 , Feb-2022